Nikhileswara Rao Sulake

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Research Interests

Generalizable and trustworthy AI for 3D medical imaging, especially CT and MRI: foundation and vision–language models learned from large-scale, high-quality annotated data, robust to multi-center domain shift, and validated for clinical deployment. Broader interests include multimodal learning that extends to ultrasound and histopathology, and human-in-the-loop annotation with multi-radiologist review.

Education

Rajiv Gandhi University of Knowledge and Technologies, Nuzvid (RGUKT) **Eluru, India**

Bachelor of Technology in Computer Science, ***CGPA: 9.11*** *August 2023 – May 2027*

- *Also completed Pre-University Course (Class XI–XII) at RGUKT Nuzvid – CGPA: 9.88 (Dec 2021 – Aug 2023)*

Certifications

DRDO and DIAT, Pune – AI/ML Professional Certification Course **Online**

Course Guide: Dr. Upasana Singh, ***Grade: 9.2*** *June 2023 – November 2023*

Experience

TANUH **Onsite (May 2026 – Present)**

Engineering Intern (Under Prof. Phaneendra K Yalavarthy) *Bengaluru, India*

- Developing an algorithm to estimate eGFR from ultrasound images and clinical information, for onsite deployment on ultrasound machines in hospitals across India; working on generalization across multi-center, pan-India datasets using inverse CNNs, Neural ODEs and related methods.
- Built **RQC**, a web-based quality-control platform for renal ultrasound datasets: structured multi-reviewer and consensus review of images and annotations, adjudication of disagreements, random and targeted-cohort sampling, automatic metrics, reviewer dashboards, and secure cached loading with no data leakage.
- Deploying our document forgery detection algorithm (2nd place nationally) with the National Health Authority for fraud detection in health reports and licensing documents (estimated savings up to INR 600 crore), and developing **Vaidyx**, an agentic assistant that prepares, supports and follows up tumor boards for doctors.

Mohamed Bin Zayed University of Artificial Intelligence (MBZUAI) **Remote (May 2026 – Present)**

Research Assistant (Under Prof. Yanding Zhao) *Abu Dhabi, UAE*

- Working on predicting whole genome duplication in cervical and ovarian cancer patients directly from H&E-stained histopathology slides using several Foundation Models with proposing MIL method.
- Working on an agentic multimodal WSI analysis pipeline that combines VLM-based navigation, pathology foundation models and WSI-level MIL for efficient, adaptive, early-stopping diagnosis.

Neural Nurture **Remote (Aug 2025 – Nov 2025)**

Research Intern - LLM *Indore, India*

- Worked on Project Lens to assess and mitigate PII and privacy leakage in large language models under white-box and black-box threat settings; systematic experiments observed 4–5% PII leakage across open-source models, highlighting potential gaps in training-data sanitization and PII removal mechanisms.

Stanford Medicine Department of Radiology **Remote (June 2025 – Aug 2025)**

Fellowship in Molecular Biology (Under Drs. Frezghi Habte and Arutselvan Natarajan) *California, USA*

- Conducted research on AI applications in medical imaging with a focus on Alzheimer's disease, including analysis of brain imaging data and presentation of findings during the fellowship.

School of Medical Science and Technology (SMST) at IIT Kharagpur **Hybrid (May 2025 – Aug 2025)**

Summer Research Intern (Under Dr. Subhamoy Mandal) *Kolkata, India*

- Developed TARU-Net, a unified framework combining U-Net-based CT lung segmentation with topology-aware 3D reconstruction using marching cubes, and applied it to real-world patient data.
- Developed a clinically applied algorithm that creates a real-time, breathing 3D lung digital twin from wearable biosignals for respiratory monitoring (accepted at MICCAI 2026).

Publications

Peer-Reviewed / Accepted

[1] **CXR-LT 2026 Challenge: Multi-Center Long-Tailed and Zero-Shot Chest X-ray Classification**
Hexin Dong, Yi Lin, Pengyu Zhou, Fengnian Zhao, Alan Clint Legasto, Juno Cho, Dohui Kim, Justin Namuk Kim, Mingeon Kim, Sunwoo Kwak, Gabriel Moyà-Alcover, Ky Trung Nguyen, Thanh-Huy Nguyen, Ha-Hieu Pham, Huy-Hieu Pham, Huy Le Pham, **Nikhileswara Rao Sulake**, Aina Tur-Serrano, Ruichi Zhang, Ang Zu, Adam E. Flanders, Zhiyong Lu, Ronald M. Summers, Mingquan Lin, Hao Chen, Yuzhe Yang, George Shih, Yifan Peng
Medical Image Analysis (IF: 14.0) [Paper Link]

[2] **Digital Twin of the Lung from Wearable Biosignals for Real-Time Respiratory Monitoring**
Partha Acharya, **Nikhileswara Rao Sulake**, Soutrik Chakraborty, Subhamoy Mandal, Suman Chakraborty
Medical Image Computing and Computer Assisted Intervention (MICCAI), Main Conference, 2026 [Paper Link]

[3] **CIPS-Net: Text-Instructed Conditional Nucleus Instance Segmentation in Histopathology**
Nikhileswara Rao Sulake*, Sai Manikanta Eswar Machara*, Sivaji Retta, Iyyakutti Iyappan Ganapathi, Naoufel Verghi
COMPAYL Workshop, MICCAI, 2026 [Paper Link]

[4] **Speckle-Aware Signal Extraction as an Alternative to Complex Methods for ECG-Free Cardiac Phase Detection**
Nikhileswara Rao Sulake, Sai Manikanta Eswar Machara, Sivaji Retta, Iyyakutti Iyappan Ganapathi, Naoufel Verghi
ASMUS Workshop, MICCAI, 2026 [Paper Link]

[5] **GOOSE-M2F: Adapting Mask2Former for High-Fidelity, Long-Tailed Fine-Grained Semantic Segmentation in Unstructured Outdoor Terrain**
Jyothiraditya Lingam, **Nikhileswara Rao Sulake**, Sai Manikanta Eswar Machara
Workshop on Field Robotics, International Conference on Robotics and Automation (ICRA), 2026 [Paper Link]

[6] **Benchmarking Input Strategies and Hybrid Loss Functions for Multi-Sequence Abdominal MRI Segmentation**
Nikhileswara Rao Sulake, Divya Katam, Sivalal Kethavath
IEEE Guwahati Subsection Conference, 2026 [Paper Link]

[7] **Uncertainty Quantification and Calibration in Retinal OCT Classification with CNN and MLP-Mixer Models**
Nikhileswara Rao Sulake, Sivalal Kethavath
IEEE Guwahati Subsection Conference, 2026 [Paper Link]

[8] **YOLOv11 Architecture Explained: Next-Level Object Detection with Enhanced Speed and Accuracy**
Nikhileswara Rao Sulake
Computer Vision Conference (CVC), Netherlands, 2026 [Paper Link]

Under Review

[1] **A Hyperbolic One-Class Fine-Tuning of Foundation Models for Robust Medical Deepfake Detection**
Saurabh Sharma*, Sai Manikanta Eswar Machara*, **Nikhileswara Rao Sulake***, Sivalal Kethavath, Phaneendra K. Yalavarthy
IEEE Transactions on Image Processing

[2] **TARU-Net: Topology-Aware Deep Learning for High-Fidelity Lung CT Segmentation and Volume Visualization**
Partha Acharya, Soutrik Chakraborty, **Nikhileswara Rao Sulake**, Aishik Sen, Amitabha Chattopadhyay, Subhamoy Mandal, Suman Chakraborty
Journal of Medical Imaging

[3] **Hybrid CNN-iSURF Feature Extraction: A Robust Approach for Image Matching Using Deep Learning and Keypoint-Based Descriptors**

Partha Acharya, Harsh Vardhan, Vineeth Bhanukoti, **Nikhileswara Rao Sulake**, Soutrik Chakraborty, Subhamoy Mandal, Suman Chakraborty
IEEE Transactions on Image Processing

[4] **Image Data Augmentation for Autonomous Vehicle Dataset using Neural Style Transfer**
Partha Acharya, Soutrik Chakraborty, **Nikhileswara Rao Sulake**, Subhamoy Mandal, Suman Chakraborty
Indian Conference on Computer Vision, Graphics and Image Processing (ICVGIP), 2026

Preprints & Other Writing

[1] **Loss Design and Architecture Selection for Long-Tailed Multi-Label Chest X-Ray Classification**
Nikhileswara Rao Sulake
arXiv Preprint, 2026 – [arXiv:2603.02294]

[2] **MINDER: Machine Learning Framework for Depression Score Analysis in Mindfulness Interventions**
Nikhileswara Rao Sulake, Sai Manikanta Eswar Machara, Divya Katam
arXiv Preprint, 2026

[3] **Survey on Deep Learning and Explainable AI for MRI-Based Alzheimer's Disease Classification**
Nikhileswara Rao Sulake, Sidharth Pancholi
Authorea Preprint, 2025 [Paper Link]

** Denotes equal contribution.*

Technical Skills

Programming Languages: Python, C

Frameworks & Libraries: PyTorch, TensorFlow, Hugging Face, CUDA

Computer Vision: OpenCV, Segmentation, Object Detection, Vision-Language Models, Multi-Task Learning

Medical AI: Medical Image Analysis (CT, MRI, Ultrasound, Histopathology), Multimodal Learning, MRI Physics

Medical Imaging Tools: MONAI, nnU-Net (incl. active learning), SimpleITK, ITK-SNAP, 3D Slicer, RadiAnt

Tools & Deployment: Git, Weights & Biases, TensorBoard, Linux, Cloud Services, On-Premise Deployment

Honors & Awards

MICCAI 2026 Travel Grant Awardee – Received MICCAI Society 2026 Travel Grant for presenting 3 MICCAI papers in Strasbourg, France. *2026*

2nd Place (National), NHA Auto Adjudication Hackathon – Developed a purely mathematical solution for real-time document forgery detection. *2026*

Top 5, CXR Long Tail Challenge (ISBI 2026) – Systematic analysis of long-tailed Chest X-Ray classification. *2026*

IEEE EMBS BHI 2025 Data Competition Finalist – Developed MINDER, a machine learning framework for BDI depression score prediction and analysis. *2025*

AI 4 Andhra Police Hackathon Winner – Built a product for digitising handwritten multi-lingual documents with a user-admin web interface. *2025*

SNU Multi-Modal AI 4 TB – Top 22 – Led sole Indian team in chest X-ray + text analysis at Seoul National University competition. *2024*

Intel AI 2024 Hackathon – 2nd Runner-Up – Developed AI-driven echocardiography analysis tools, presented at IIT Kharagpur (IEEE Indicon 2024). *2024*